

# Natural Units, the Cosmological Hierarchy, and a Ramanujan–Heegner Numerical Curiosity

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Buckingham’s  $\Pi$ -theorem provides a linear-algebraic account of natural units. The constants  $c$ ,  $\hbar$ , and  $G$  form a dimensionally independent basis for the mechanical dimensions  $M$ ,  $L$ , and  $T$ . Introducing the Hubble parameter  $H$  produces one dimensionless invariant,  $\hbar G H^2 / c^5$ . Depending on the chosen natural-unit basis, the observed hierarchy of order  $10^{-122}$  may appear in  $H^2$ ,  $G$ , or  $\Lambda \ell_P^2$ . A numerical curiosity then follows: the inverse seventh power of the Ramanujan constant  $\mathcal{R} = e^{\pi\sqrt{163}}$  is also approximately  $10^{-122}$ . The numbers 7 and 163 are both Heegner numbers, though no physical mechanism is implied.

## I. BUCKINGHAM’S THEOREM

Buckingham’s  $\Pi$ -theorem states that a physical relation involving  $n$  dimensional quantities constructed from  $r$  independent base dimensions may be expressed using  $n - r$  independent dimensionless combinations.

For relativistic gravitation, take the mechanical base dimensions to be  $[M]$ ,  $[L]$ ,  $[T]$ . Represent each physical quantity by its exponent vector in the ordered basis  $(M, L, T)$ . The dimensions of the speed of light, the reduced Planck constant, and Newton’s gravitational constant are

$$\begin{aligned} [c] &= LT^{-1}, \\ [\hbar] &= ML^2T^{-1}, \\ [G] &= M^{-1}L^3T^{-2}. \end{aligned}$$

Their dimensional exponent vectors are therefore

$$d_c = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad d_\hbar = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad d_G = \begin{pmatrix} -1 \\ 3 \\ -2 \end{pmatrix}.$$

Using these vectors as columns gives the dimensional matrix

$$D = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 2 & 3 \\ -1 & -1 & -2 \end{pmatrix}.$$

Its determinant is  $\det D = -2 \neq 0$ . Hence  $\text{rank}(D) = 3$ , and the three constants  $c$ ,  $\hbar$ , and  $G$  are dimensionally independent. They form a basis for the three-dimensional mechanical dimensional space. This establishes the linear-algebraic legitimacy of the natural-unit choice  $c = \hbar = G = 1$ . The constants have not disappeared physically. Their dimensions have been absorbed into the definitions of the units themselves.

## II. ADDING THE HUBBLE PARAMETER

The Hubble parameter has dimensions  $[H] = T^{-1}$ , and hence dimensional vector  $d_H = (0 \ 0 \ -1)^T$ . Adjoining this fourth quantity gives

$$D_{\text{ext}} = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 1 & 2 & 3 & 0 \\ -1 & -1 & -2 & -1 \end{pmatrix}.$$

The rank remains three:  $\text{rank}(D_{\text{ext}}) = 3$ . Since there are four quantities and three independent base dimensions,  $\text{nullity}(D_{\text{ext}}) = 1$ . Buckingham’s theorem therefore guarantees one independent dimensionless combination. Let  $\Pi = c^a \hbar^b G^d H^e$ , so dimensional homogeneity amounts to

$$D_{\text{ext}} \begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} a \\ b \\ d \\ e \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \\ 1 \\ 2 \end{pmatrix}.$$

The resulting invariant is

$$\Pi = \frac{\hbar G H^2}{c^5}.$$

Using the Planck time,

$$t_P = \sqrt{\frac{\hbar G}{c^5}},$$

this may be written  $\Pi = (H t_P)^2$ . For the present Hubble parameter  $H_0$ ,  $H_0 t_P \sim 10^{-61}$ , and therefore

$$\Pi_0 = \frac{\hbar G H_0^2}{c^5} \sim 10^{-122}. \quad (1)$$

This small dimensionless number cannot be eliminated through a change of units. A change of basis can only determine which dimensional constant carries it. In Planck units one chooses  $c = \hbar = G = 1$ , and the invariant then reduces to  $\Pi_0 = H_0^2$ , so that  $H_0^2 \sim 10^{-122}$ . One may instead adopt a Hubble-adapted basis,  $c = \hbar = H_0 = 1$ , so invariant becomes  $\Pi_0 = G$ , and therefore

$$G_{\{c=\hbar=H_0=1\}} \sim 10^{-122}.$$

Same hierarchy expressed in another dimensional basis.

## III. COSMOLOGICAL CONSTANT

Consider de Sitter spacetime and Planck length,

$$H^2 = \frac{\Lambda c^2}{3}, \quad \ell_P = \sqrt{\frac{\hbar G}{c^3}}.$$

Multiplying the de Sitter relation by  $\hbar G/c^5$  gives

$$\frac{\hbar G H^2}{c^5} = \frac{\Lambda \hbar G}{3c^3}.$$

Since  $\ell_p^2 = \frac{\hbar G}{c^3}$ , it follows that

$$\Lambda \ell_p^2 = 3 \frac{\hbar G H^2}{c^5}.$$

Thus the same hierarchy may be expressed as

$$\Lambda \ell_p^2 \sim 10^{-122}.$$

#### IV. A RAMANUJAN-HEEGNER COINCIDENCE

Define the Ramanujan constant  $\mathcal{R} = e^{\pi\sqrt{163}}$ . Numerically,  $\mathcal{R} \approx 2.6253741264 \times 10^{17}$ . Its seventh inverse power is

$$\mathcal{R}^{-7} = e^{-7\pi\sqrt{163}} \approx 1.1632 \times 10^{-122}$$

In Hubble-adapted natural units, we have numerical comparison

$$G_{\{c=\hbar=H_0=1\}} \sim 10^{-122} \approx \mathcal{R}^{-7}.$$

The pair (7, 163) is arithmetically conspicuous because both entries are Heegner numbers. The complete set of Heegner numbers is

$$1, 2, 3, 7, 11, 19, 43, 67, 163.$$

The appearance of 163 in  $\mathcal{R} = e^{\pi\sqrt{163}}$  is tied to the class-number-one property of the imaginary quadratic field

$$\mathbb{Q}(\sqrt{-163})$$

and to the theory of complex multiplication of elliptic curves. The exponent 7, though itself a Heegner number, has no established cosmological derivation in this setting.

Consider now the even Riemann zeta value

$$\zeta(6) = \sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945} \approx 1.017343062.$$

Define the adjusted quantity  $\mathcal{R}' = \frac{\mathcal{R}}{\zeta(6)}$ , so we have,

$$\frac{1}{(\mathcal{R}')^7} = \left( \frac{\zeta(6)}{\mathcal{R}} \right)^7 = \zeta(6)^7 e^{-7\pi\sqrt{163}} \approx 1.31200 \times 10^{-122}.$$

Euler's Basel evaluation,

$$\zeta(2) = \frac{\pi^2}{6}$$

allows us to write  $\pi = \sqrt{6\zeta(2)}$ , so  $\pi\sqrt{163} = \sqrt{978\zeta(2)}$  and the Ramanujan constant may be written without an explicit  $\pi$  as  $\mathcal{R} = \exp(\sqrt{978\zeta(2)})$ . The adjusted seventh inverse power becomes

$$\left( \frac{\exp(\sqrt{978\zeta(2)})}{\zeta(6)} \right)^{-7} \approx 1.31200 \times 10^{-122}.$$

Of course we can play this game all we like, noting the numerology lost as adjusted  $\mathcal{R}$  loses its mythical integer status.

TABLE I. Zeta-adjusted Ramanujan values,  $\mathcal{R}'_s = \mathcal{R}/\zeta(s)$ .

| $s$ | $\zeta(s)$  | $(\mathcal{R}'_s)^{-7}$    |
|-----|-------------|----------------------------|
| 4   | 1.082323234 | $2.02377 \times 10^{-122}$ |
| 5   | 1.036927755 | $1.49934 \times 10^{-122}$ |
| 6   | 1.017343062 | $1.31200 \times 10^{-122}$ |

#### Appendix A: Error Analysis

The quoted  $10^{-122}$  is an order-of-magnitude statement. Since  $G_H = \hbar G H_0^2/c^5 \propto H_0^2$ , the fractional uncertainty is

$$\frac{\Delta G_H}{G_H} = 2 \frac{\Delta H_0}{H_0}.$$

Using 1 Mpc =  $3.08567758 \times 10^{19}$  km,

$$H_0(\text{s}^{-1}) = \frac{H_0(\text{km s}^{-1} \text{Mpc}^{-1})}{3.08567758 \times 10^{19}}.$$

Thus  $H_0^{\text{CMB}} = 67.4 \pm 0.5$  gives  $H_0^{\text{CMB}} \approx 2.1843 \times 10^{-18} \text{ s}^{-1}$ , while  $H_0^{\text{local}} = 73.04 \pm 1.04$  gives  $H_0^{\text{local}} \approx 2.3679 \times 10^{-18} \text{ s}^{-1}$ . Using  $\hbar = 1.054571817 \times 10^{-34} \text{ Js}$ ,  $G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ ,  $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$ , Eq. (1) gives

$$G_H^{\text{CMB}} = (1.3868 \pm 0.0206) \times 10^{-122}$$

$$G_H^{\text{local}} = (1.6285 \pm 0.0464) \times 10^{-122}.$$

The relative uncertainties are 1.48% and 2.85%, respectively, so the spread between the two central values is larger than the uncertainty within either measurement. Indeed, the midpoint between the two central estimates is

$$\frac{G_H^{\text{CMB}} + G_H^{\text{local}}}{2} = 1.50765 \times 10^{-122},$$

which lies strikingly close to the  $\zeta(5)$ -adjusted Ramanujan value, being only 0.55% below the midpoint. Something to aim for in the absence of discernible empiricism.